

Low-Level Sensor Signal Conditioning Circuit Design

CIE 201 — Advanced Electric Circuits | Phase 2 Report | Spring 2026

Zewail City of Science and Technology

Mohammed Soliman, Mohamed Tamer, Omar Ihab, Ahmed Akef, Karim Assem

202402280, 202400792, 202401437, 202401486, 202400688

1. DETAILED STAGE DESIGN

The signal conditioning circuit processes the K-Type thermocouple output through four sequential stages, each designed and verified independently before integration into the complete system.

1.1 Stage 1 — K-Type Thermocouple

The thermocouple output is modeled as a superposition of a slow-varying DC component and a 50 Hz interference term:

$$V_{\text{sensor}} = V_{\text{DC}} + V_{\text{noise}}$$

where the DC component is given by the Seebeck relationship:

$$V_{\text{DC}} = 41 \mu\text{V}/^\circ\text{C} \times (T_{\text{hot}} - T_{\text{cold}})$$

At a representative temperature difference of $\Delta T = 25^\circ\text{C}$:

$$V_{\text{DC}} = 41 \times 25 = 1025 \mu\text{V} \approx 1 \text{ mV}$$

1.2 Stage 2 — Instrumentation Amplifier

Two MCP6002 rail-to-rail op-amps are configured as a three-op-amp instrumentation amplifier to amplify the weak differential thermocouple signal while rejecting common-mode interference.

1.2.1 Gain Calculation

The differential voltage gain is set by resistors R3, R4, and R5:

$$A_v = 1 + 2R3/R5 = 1 + (2 \times 500 \text{ k}\Omega)/(1 \text{ k}\Omega) = 1001$$

The resulting amplified output voltage is:

$$V_{\text{out}} = 1.025 \text{ mV} \times 1001 \approx 1.026 \text{ V}$$

1.3 Stage 3 — Second-Order Low-Pass Filter

A cascaded two-stage passive RC filter eliminates the 50 Hz power-line interference. Each stage is designed with the standard first-order cutoff frequency formula:

$$f_c = 1 / (2\pi RC)$$

The first stage ($R1 = 1 \text{ k}\Omega$, $C1 = 100 \mu\text{F}$) yields a cutoff of:

$$f_{c1} = 1 / (2\pi \times 1 \text{ k}\Omega \times 100 \mu\text{F}) \approx 1.59 \text{ Hz}$$

The second stage ($R2 = 10 \text{ k}\Omega$, $C2 = 1000 \mu\text{F}$) provides deeper attenuation at an even lower cutoff:

$$f_{c2} = 1 / (2\pi \times 10 \text{ k}\Omega \times 1000 \mu\text{F}) \approx 0.016 \text{ Hz}$$

The combined two-stage response ensures complete rejection of the 50 Hz interference while preserving the DC temperature signal.

1.4 Stage 4 — Output Scaling Amplifier

A non-inverting amplifier scales the filtered $\approx 1 \text{ V}$ signal up to the required 0–5 V output range. The gain is set by resistors Rf and Rin:

$$A_v = 1 + R_f/R_{in} = 1 + 40 \text{ k}\Omega/10 \text{ k}\Omega = 5$$

This maps the $\approx 1.026 \text{ V}$ filtered output to $\approx 5.13 \text{ V}$, which is then clamped to 5 V by the supply rail, giving the final 0–5 V output.

2. COMPONENT LIST

Table 1: Main Components

#	Component	Specification	Qty
1	MCP6002	Rail-to-rail Op-Amp	3
2	Resistors	1 k Ω – 500 k Ω	Multiple
3	Capacitors	100 μF and 1000 μF	2
4	Breadboard	830 tie-point	1
5	Power Supply	5 V regulated	1

3. IMPLEMENTATION DOCUMENTATION

The complete circuit was assembled on an 830 tie-point breadboard and tested stage-by-stage before integration. Each stage was verified independently using a digital multimeter and oscilloscope before connecting it to the next.

4. WAVEFORM DOCUMENTATION

Oscilloscope captures were recorded at each stage of the signal conditioning chain to verify correct operation. The sequence of waveforms below confirms that the signal is successfully amplified, filtered, and scaled at each stage.

5. DESIGN SUMMARY

Table 2: Signal Levels Across Stages

Stage	Block	Input	Output
1	Thermocouple	Temp. difference	$\sim 1 \text{ mV} + \text{noise}$
2	Instrumentation Amp.	$\sim 1 \text{ mV}$	$\sim 1 \text{ V}$
3	Low-Pass Filter	Signal + noise	Clean DC
4	Scaling Amplifier	$\sim 1 \text{ V}$	0–5 V

Design the Circuit on Simulink

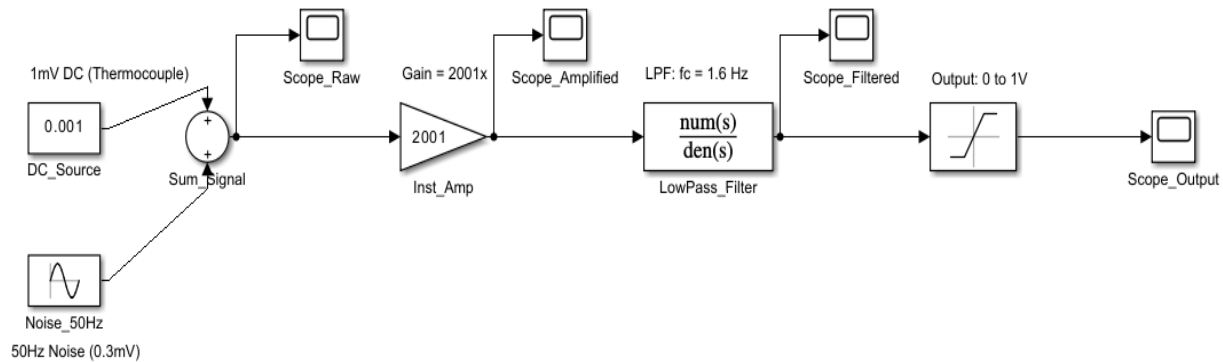


Figure 1 : Simulink (Block Diagram)

Design the filter (Second Order)

Design and Implementation of a 2nd-Order Low-Pass Filter

To suppress the 50 Hz powerline interference while preserving the low-frequency thermocouple signal, a second-order Butterworth low-pass filter was designed. Compared to a first-order filter, the second-order response provides a steeper attenuation rate of 40 dB/decade, which improves noise rejection performance.

Design Parameters

The filter was designed using the following specifications:

- Cutoff frequency: $f_c = 1.6$ Hz
- Damping ratio for Butterworth response: $D = 0.707$
- Filter type: Second-Order Continuous-Time Low-Pass Filter

Angular Cutoff Frequency

The angular cutoff frequency was calculated using:

$$\begin{aligned}\omega_c &= 2\pi f_c \\ \omega_c &= 2 \times \pi \times 1.6 \\ \omega_c &\approx 10.0531 \text{ rad/s}\end{aligned}$$

Transfer Function Derivation

The general transfer function of a second-order low-pass filter is given by:

$$H(s) = \omega_c^2 / (s^2 + 2D\omega_c s + \omega_c^2)$$

For a Butterworth response:

$$2D\omega_c = \sqrt{2} \omega_c$$

Using $\omega_c \approx 10.0531$, the coefficients are calculated as follows.

1. Numerator Term

$$\begin{aligned}\omega_c^2 &= (10.0531)^2 \\ \omega_c^2 &\approx 101.0648\end{aligned}$$

2. Coefficient of s

$$\begin{aligned}\sqrt{2}\omega_c &= \sqrt{2} \times 10.0531 \\ \sqrt{2}\omega_c &\approx 14.2172\end{aligned}$$

3. Constant Term

$$\omega c^2 \approx 101.0648$$

Therefore, the final transfer function becomes:

$$H(s) = 101.0648 / (s^2 + 14.2172s + 101.0648)$$

Simulink Implementation

The filter was implemented in Simulink using the Transfer Function block with the following coefficient vectors:

- Numerator coefficients: [101.0648]
- Denominator coefficients: [1 14.2172 101.0648]

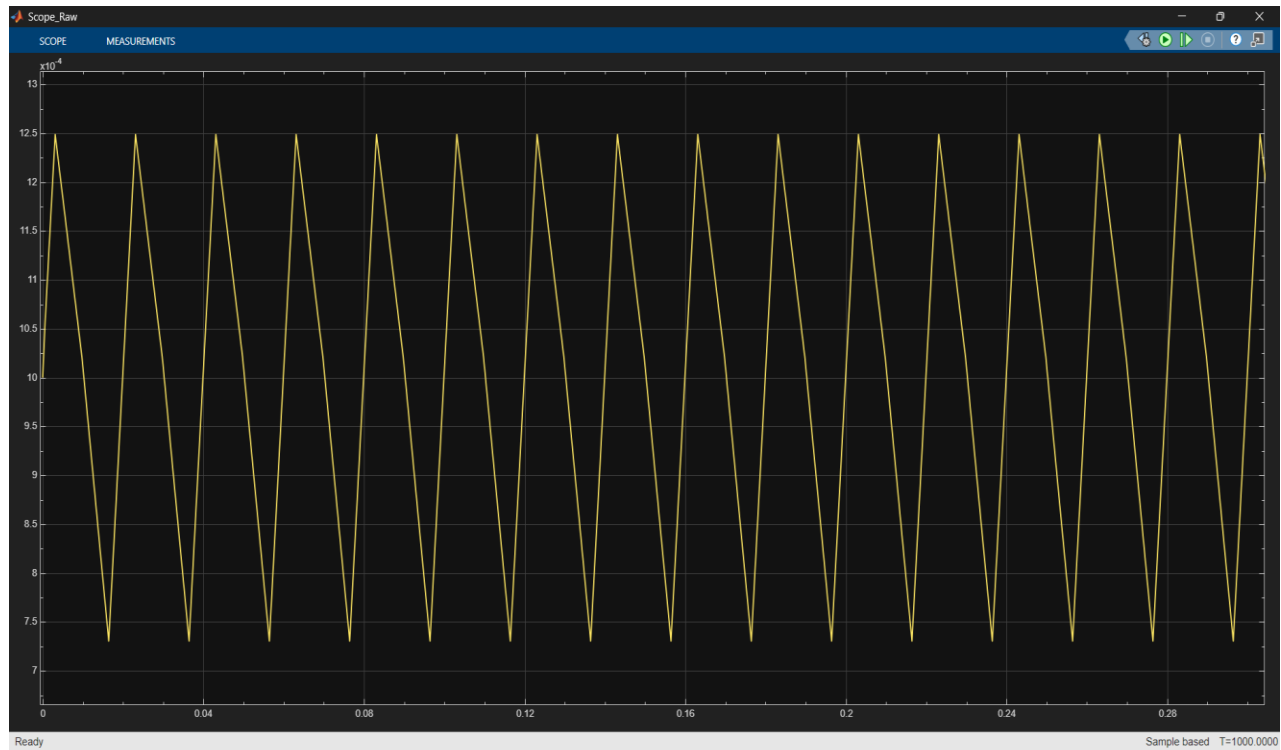


Figure 1 : Before Amplification

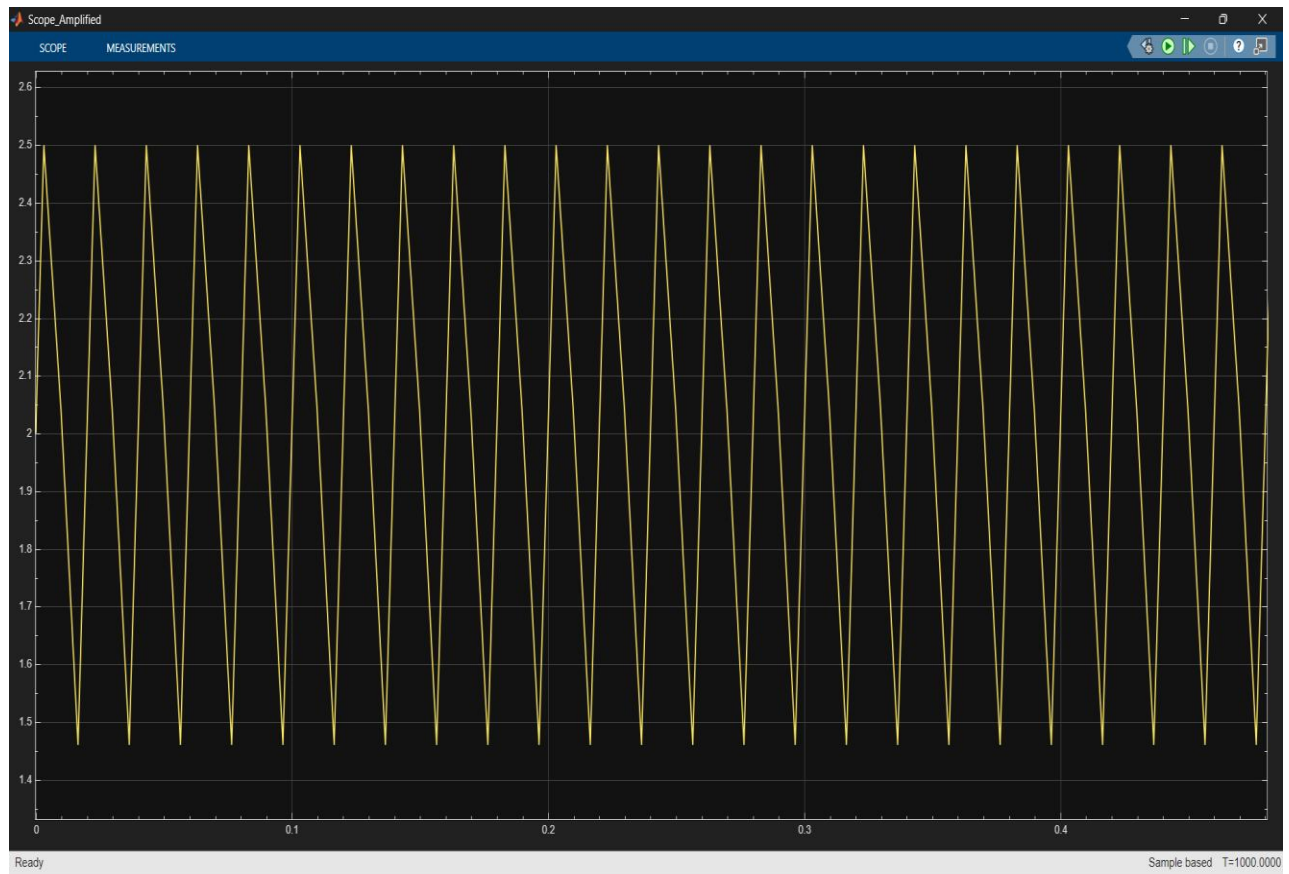


Figure 2 : After Amplification

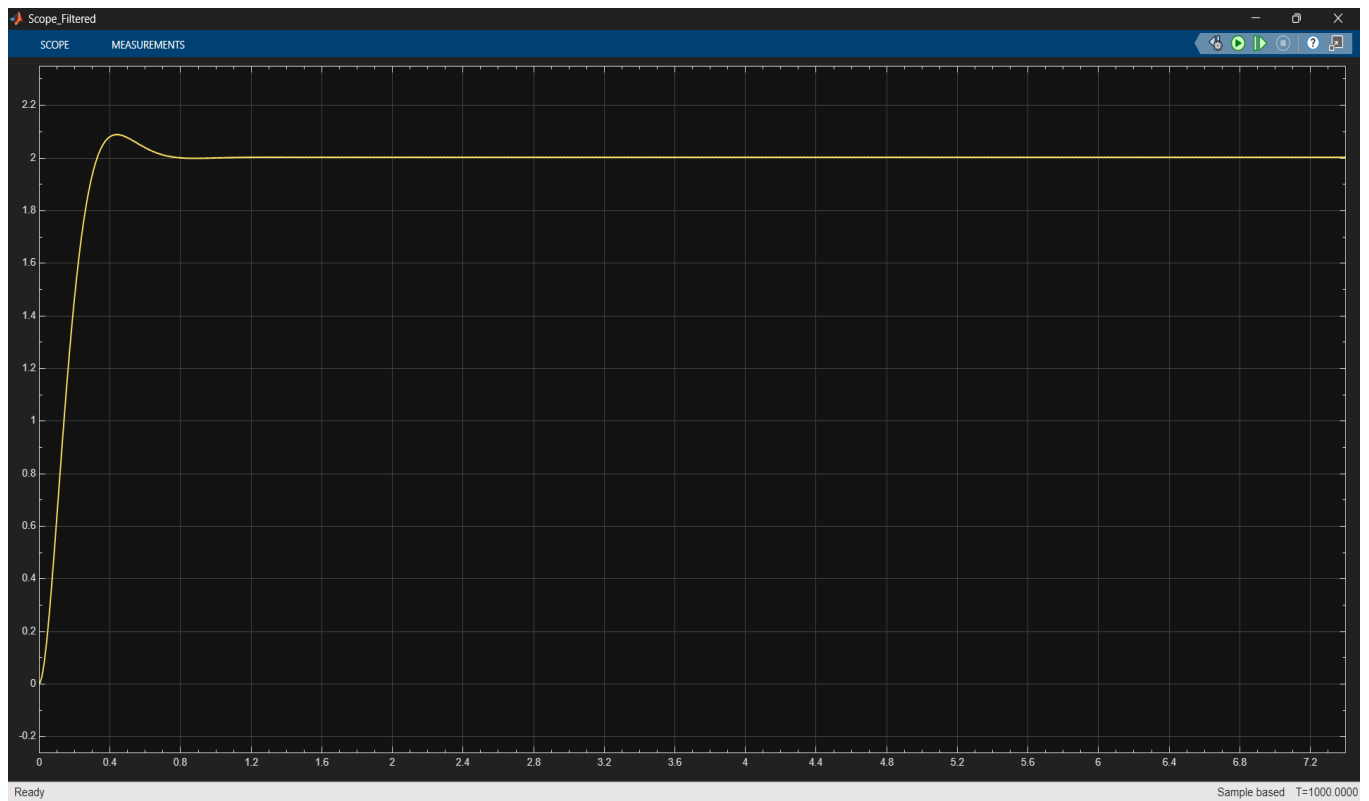


Figure 3 : After Amplification and Applying Filter

